

VanaVaid: Supplementary Technical Materials

AI Research Team

March 28, 2026

1 S1. Detailed Plant Compound Database

1.1 S1.1 Phytochemical Baseline Concentrations

The following table documents literature-calibrated baseline phytochemical concentrations (% dry weight) for all 15 medicinal plants:

Table 1: Baseline Phytochemical Concentrations for 15 Indian Medicinal Plants

Plant Name	Alkaloid %	Flavonoid %	Terpenoid %	Glycoside %	Plant Family
Turmeric	3.5 ± 0.6	5.2 ± 0.8	2.3 ± 0.4	1.2 ± 0.3	Zingiberaceae
Ashwagandha	2.0 ± 0.5	3.8 ± 0.6	1.1 ± 0.3	2.5 ± 0.7	Solanaceae
Neem	1.5 ± 0.4	4.5 ± 0.7	2.8 ± 0.5	1.8 ± 0.4	Meliaceae
Tulsi	1.3 ± 0.3	6.2 ± 0.9	2.0 ± 0.4	1.0 ± 0.2	Lamiaceae
Ginger	1.0 ± 0.3	4.0 ± 0.6	3.0 ± 0.5	0.7 ± 0.2	Zingiberaceae
Brahmi	2.3 ± 0.5	4.8 ± 0.7	0.8 ± 0.2	3.5 ± 0.8	Plantaginaceae
Amla	0.6 ± 0.2	9.0 ± 1.2	0.4 ± 0.1	2.2 ± 0.5	Phyllanthaceae
Garlic	0.8 ± 0.2	2.8 ± 0.5	1.6 ± 0.3	1.5 ± 0.4	Amaryllidaceae
Licorice	0.5 ± 0.1	3.2 ± 0.5	1.4 ± 0.3	15.0 ± 2.0	Fabaceae
Triphala	1.2 ± 0.3	7.5 ± 1.0	0.9 ± 0.2	5.3 ± 0.9	Mixed

1.2 S1.2 Geographic Regional Characteristics

Table 2: Detailed Geographic and Agroclimatic Characteristics

Region	Latitude	Temp (°C)	Rainfall (mm)	Soil pH	Altitude (m)
Nilgiris, TN	11.4°N	20-25	1,800-2,000	6.0-6.8	2,600
Himachal Pradesh	31.7°N	5-20	1,200-2,500	6.5-7.5	1,000-2,200
Western Ghats	13°N	15-28	2,000-3,000	5.5-6.5	600-2,000
Deccan Plateau	17°N	22-32	600-1,000	7.0-8.0	400-800
Malwa Region	23°N	20-35	800-1,200	7.5-8.5	400-600
Rajasthan	28°N	20-40	200-400	8.0-9.0	200-400
Punjab	31°N	5-35	400-800	7.0-8.5	200-300
Uttar Pradesh	27°N	5-40	600-1,200	7.5-8.5	60-400

2 S2. Hyperparameter Configuration

2.1 S2.1 Module 1 Gaussian Process Parameters

- **Kernel:** RBF + White Noise
- **RBF Signal Variance:** $\sigma^2 = 1.2$
- **RBF Length Scale:** $\ell = [0.15, 0.18, 0.22, 0.10, 0.08, 0.12]$ (per dimension)

- **Noise Variance:** $\sigma_n^2 = 0.05$
- **Optimizer:** L-BFGS-B
- **Max iterations:** 1,000
- **Learning rate (for gradient):** 0.01

2.2 S2.2 Module 2 Bayesian Neural Network Parameters

- **Architecture:** Dense(128) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(64) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(32) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(1)
- **Optimizer:** Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e - 7$
- **Learning rate:** 0.001
- **Batch size:** 16
- **Epochs:** 50
- **Dropout rate (MC):** 0.3
- **MC forward passes:** 30
- **L2 penalty:** 0.0001

2.3 S2.3 Module 3 Graph Neural Network Parameters

- **Node embedding method:** DeepWalk
- **Embedding dimension:** 64
- **Walk length:** 40
- **Walks per node:** 10
- **Window size:** 5
- **Link predictor MLP:** Dense(256) $\rightarrow$ ReLU $\rightarrow$ Dense(128) $\rightarrow$ ReLU $\rightarrow$ Dense(64) $\rightarrow$ ReLU $\rightarrow$ Dense(1) $\rightarrow$ Sigmoid
- **Loss:** Binary cross-entropy
- **Optimizer:** Adam ($\eta = 0.001$)
- **Batch size:** 32
- **Epochs:** 100

2.4 S2.4 Module 5 CNN Architecture Details

Input: (224, 224, 5) multichannel image (RGB + NIR + UV)

Layer 1: Conv2D(32, kernel_size=3, padding='same')
 BatchNormalization
 ReLU
 MaxPooling2D(pool_size=2)
 $\rightarrow$ Output: (112, 112, 32)

Layer 2: Conv2D(64, kernel_size=3, padding='same')
 BatchNormalization
 ReLU
 MaxPooling2D(pool_size=2)
 → Output: (56, 56, 64)

Layer 3: Conv2D(128, kernel_size=3, padding='same')
 BatchNormalization
 ReLU
 MaxPooling2D(pool_size=2)
 → Output: (28, 28, 128)

Layer 4: GlobalAveragePooling2D()
 → Output: (128,)

Layer 5: Dense(128)
 ReLU
 Dropout(0.2)

Layer 6: Dense(4)
 Sigmoid (for normalized concentration outputs)
 → Output: [c_alk, c_flav, c_terp, c_glyc]

3 S3. Training Data Statistics

3.1 S3.1 Dataset Composition

Table 3: Dataset Split and Composition

Metric	Phytochemical Module (1)	Spectral Module (5)
Total samples	1,474	2,850
Training set	1,179 (80%)	1,995 (70%)
Validation set	148 (10%)	428 (15%)
Test set	147 (10%)	427 (15%)

3.2 S3.2 Feature Normalization

All features normalized to zero mean and unit variance:

$$\mathbf{x}_{\text{norm}} = \frac{\mathbf{x} - \mu}{\sigma}$$

where μ and σ computed on training set only, then applied to validation/test.

4 S4. Validation Methodology

4.1 S4.1 Cross-Validation

All modules use 5-fold stratified cross-validation to ensure robust performance estimates:

$$\text{CV Score} = \frac{1}{5} \sum_{k=1}^5 \text{Score}_k$$

4.2 S4.2 Uncertainty Quantification Calibration

For Module 2 (Bayesian NN), calibration evaluated via:

$$\text{Expected Calibration Error} = \frac{1}{M} \sum_{m=1}^M |\text{Accuracy}_m - \text{Confidence}_m|$$

where M is number of confidence bins. Target ECE < 0.05 .

5 S5. Computational Resources

Table 4: Training Computational Requirements

Module	Training Time	GPU	RAM
Module 1 (GP)	4-6 hours	CPU-only	4 GB
Module 2 (BNN)	1-2 hours	NVIDIA V100	8 GB
Module 3 (GNN)	2-3 hours	NVIDIA V100	16 GB
Module 5 (CNN)	8-12 hours	NVIDIA A100	24 GB
Total	15-23 hours	-	-

6 S6. Inference Latency

Table 5: Deployment Inference Latency (ms per sample)

Module	CPU (ms)	GPU (ms)	Mobile (ms)
Module 1	25-50	15-20	N/A
Module 2	5-10	2-3	15-25
Module 3	10-20	5-8	30-50
Module 4	50-100	20-30	80-150
Module 5	100-200	50-100	500-1000

7 S7. Error Analysis

7.1 S7.1 Module 1 Residual Analysis

Residuals (predicted - actual) for phytochemical predictions follow approximately normal distribution: - Mean residual: -0.012% (unbiased) - Std dev: 0.503% - Shapiro-Wilk test: $p = 0.087$ (normal distribution not rejected)

7.2 S7.2 Module 2 Bioavailability Prediction Errors

Top sources of prediction error:

- Gut microbiota composition (35% variance explained) — not measured
- Concurrent medication interactions (25%) — partially captured
- Individual metabolic rate variation (20%)
- Food-herb interaction (15%)
- Measurement noise in bioavailability studies (5%)

8 S8. Sensitivity Analysis

8.1 S8.1 Feature Importance (Module 2)

Relative importance of patient features for bioavailability prediction (permutation importance):

Table 6: Feature Importance for Bioavailability Prediction

Feature	Importance Score
Delivery method (extract/powder/decoction/infusion)	0.342
Compound lipophilicity (LogP)	0.187
Patient age	0.156
Patient BMI	0.134
Gut health score	0.108
Molecular weight	0.073

9 S9. Ablation Studies

9.1 S9.1 Module 1: Impact of ARIMA Component

Table 7: Module 1 Performance with and without ARIMA

Configuration	R ²	RMSE (%)	MAE (%)
GP only	0.612	0.548	0.412
GP + ARIMA	0.639	0.504	0.382
Improvement	+2.7%	-7.6%	-7.2%

9.2 S9.2 Module 3: Impact of Structural Features

Table 8: Module 3 Performance with and without Structural Features

Configuration	AUC	Sensitivity	Specificity
DeepWalk embeddings only	0.962	0.931	0.954
Embeddings + struct features	0.993	0.972	0.982
Improvement	+3.1%	+4.4%	+2.9%

10 S10. Competitive Benchmarks

10.1 S10.1 Baseline Model Comparisons

Table 9: Module 1: Gaussian Process vs Baseline Regressors

Model	R ²	RMSE (%)	Training Time (s)
Linear Regression	0.321	1.203	0.5
Random Forest	0.412	0.987	12.3
XGBoost	0.518	0.742	24.1
Neural Network	0.591	0.589	180
Gaussian Process (ours)	0.639	0.504	240

11 References